Research Mindmap

Gordon Pipa · Self-Organizing Recurrent Spiking Networks Research positioning within the emerging landscape of arXiv publications

1. Your Research Core

Your central research focus: self-organizing recurrent spiking neural networks. This encompasses the SORN model (2009), transfer entropy (2011), neural synchrony (2009), cross-frequency coupling (2015), and related work spanning over 50 publications with 3,735+ citations (h-index: 33).

2. Your Publications

Self-Organizing NN (SORN)	**2009**	**335**	**Recurrent / Self-org / Plasticity / Criticality**
Transfer Entropy	2011	1367	Effective Connectivity / Info Theory
Neural Synchrony	2009	1047	Oscillations / Recurrent
Cross-Frequency Coupling	2015	729	Oscillations / Network Science
Dynamical Relaying	2008	431	Recurrent / Attractor
Fading Memory	2007	109	Memory / Recurrent
NeuroXidence	2008	107	Oscillations / Network Science
Network Topology	2011	153	Network Science / Spiking
High-Gamma Activity	2011	254	Oscillations / Info Processing
Moral Decisions in VR	2014	38	VR Experiments / Info Processing

3. Research Domains

Key domains connecting your work: • Recurrent Network Dynamics — self-organization, attractor states, memory • Spiking Neural Networks — oscillations, information processing, plasticity • Information Processing — memory, effective connectivity, predictive coding • Criticality & Neuronal Avalanches — attractor dynamics, state-space • Neural Oscillations & Rhythms — synchrony, cross-frequency, effective connectivity • Self-Organization & Homeostasis — plasticity, recurrent dynamics • Network Science — high-order functional connectivity, topology Methods: Information Theory, Computational Modeling, EEG/MEG Analysis, State-Space Dynamics, Virtual Reality Experiments

4. New arXiv Papers (October 2026)

chastic Dynamics of Motif-Embedded Spiking Networks	arXiv:2610.00616	Criticality / Spiking / Network Science
ti-Scale Temporal Flows for Trajectory Generation	arXiv:2610.01086	Memory / Recurrent
dient-based STDP Solution to Credit Assignment	arXiv:2609.08070	Plasticity / Spiking / Recurrent
erminants of Hyperparameter Robustness in Reservoir Computing	arXiv:2609.07355	Recurrent / Spiking / Comp. Modeling
mation of Structural Attractors in Neuromorphic Systems	arXiv:2609.06826	Attractor / Plasticity / Spiking
ergence of Criticality in Models of Real Neurons	arXiv:2609.09438	Criticality / Comp. Modeling
ner Information Metric as Measure of Criticality	arXiv:2609.07624	Criticality / Info Theory
spective Coding Improves Deep RTN Learning	arXiv:2609.04134	Memory / Recurrent / Comp. Modeling
h-Order Triadic Functional Connectivity	arXiv:2609.03987	Network Science / Eff. Connectivity / Oscillati
nal Delay Dispersion Decides Event vs Sequence	arXiv:2609.04195	Oscillations / Info Processing / Spiking
valanches: Python Package for Spatiotemporal Avalanches	arXiv:2609.11530	Criticality / Spiking
meostasis via HMM Control	arXiv:2609.07508	Self-org / Plasticity
G Foundation Models	arXiv:2609.08694	EEG/MEG / Oscillations
nserved Hemispherical Dynamics	arXiv:2609.04308	Oscillations / Network Science

5. How New Papers Position Relative to Your Work

Criticality: 4 new papers (pyAvalanches, Fisher metric, emergence in real neurons, motif-embedded networks) extend your criticality/SORN lineage — providing new tools (pyAvalanches), new metrics (Fisher information), and new biological validation.

Plasticity & Learning: 3 papers (gradient-based STDP, structural attractors, homeostasis via HMMs) offer computational approaches that complement your SORN plasticity framework. The gradient-based STDP is particularly relevant — it bridges classical SNN learning rules with modern differentiable approaches.

Information Processing: 4 papers (prospective coding, axonal delay, high-order connectivity, Fisher metric) advance the information-theoretic perspective you established with transfer entropy. Prospective coding in RTN networks extends your fading memory work.

Oscillations: 3 papers (MEG foundation models, conserved hemispherical dynamics, axonal delay) connect your neural synchrony work to modern neuroimaging and network science approaches.

Recurrent Networks: 4 papers (temporal flows, gradient-based STDP, reservoir computing, prospective coding) push forward your recurrent dynamics research with new computational frameworks and biological grounding.

6. Key Takeaways

1. Criticality research is booming — 4 new papers suggest your work on neuronal avalanches and criticality in SORN is highly timely. The pyAvalanches package gives you a new tool. 2. Plasticity is evolving beyond biological models — gradient-based STDP and homeostatic HMM control offer new mathematical frameworks that complement your SORN plasticity rules. 3. Information processing from a network-science angle is maturing — high-order functional connectivity (triadic) and prospective coding extend your transfer entropy and fading memory work. 4. Oscillations are connecting to modern neuroimaging — MEG foundation models and conserved dynamics provide new biological validation channels for your neural synchrony research. 5. Your SORN framework positions you well — most new work validates or extends SORN principles rather than challenging them. Your work on self-organization, criticality, and plasticity forms the theoretical backbone that newer computational papers are building upon.